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Quantitative Predictions from Observational Incompleteness

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Abstract

The Observational Incompleteness (OI) framework derives the Standard Model as the unique emergent description available to observers embedded in a deterministic system on a $d = 3$ cubic lattice with spacing $2\,l_P$. We present twenty-one quantitative predictions with zero free parameters. The three gauge couplings at M_Z are reproduced to $< 0.1\%$ each. The Cabibbo angle $\lambda = 1/(\pi\sqrt{2}) = 0.22508$ matches the observed 0.22500 ± 0.00067 to 0.04% — the inverse of the Brillouin zone distance between adjacent generation corners. The Wolfenstein parameter $A = \sqrt{2/3} = 0.8165$ matches 0.826 ± 0.012 to 0.8σ — the sine of the angle between a generation axis and the Higgs direction. The Koide angle $\theta_0 = 2/9$ predicts m_e and m_μ from m_τ to better than 0.01%. Tribimaximal neutrino mixing from $A_4 \subset O$ with Cabibbo corrections gives all three PMNS angles within 0.5σ. The composite Higgs (A_1 taste) predicts $\lambda(M_{\rm Pl}) = 0$, consistent with the observed near-criticality at 0.6σ. Monte Carlo simulations at $L = 16$–64 confirm chiral condensate formation; the matching mass $m_{\rm match} = \lambda g_0^2 = 0.122$ gives $m_b/m_\tau = 4.28/Z_S = 2.36$ (observed: 2.352, 0.4%). The down-quark Koide parameter $Q_{\rm down} = (2/3)(1 + \alpha_s/\pi)$ predicts m_b from m_s to 0.9%; combined with the other relations, a single mass input (m_s) determines six fermion masses (m_d, m_u, m_b, m_τ, m_μ, m_e), all within 1%; the top mass $m_t = v/\sqrt{2}$ follows from the IR quasi-fixed point $y_t = 1$ (0.9%). All structural constants trace to the geometry of $d = 3$: $\pi\sqrt{2}$, $\sqrt{2/3}$, 2/9, $\sqrt{2/9}$, 2/3, 4/9, 1/3, 1/2.

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1. Introduction

The Standard Model contains approximately 19 free parameters, none derived from a deeper principle. The Observational Incompleteness (OI) framework [1, 2] proposes that these parameters are consequences of embedded observation: the SM is the unique quantum field theory that arises when an observer traces out inaccessible degrees of freedom in a deterministic system.

The framework begins with a single empirical fact — *observation occurs* — formalized as a definition: an observation is a triple (S, φ, V) consisting of a total system, a deterministic dynamics, and an embedded observer [1]. Three structural lemmas follow: finiteness of the visible sector, causal partition into visible and hidden sectors, and the unique invariant measure. The emergent

description on V is unitarily evolving quantum mechanics [1], and the $d = 3$ cubic lattice structure induces the SM gauge group SU(3) × SU(2) × U(1) with specific coupling constants [2].

We present twenty-one predictions spanning gauge couplings, quark mixing, lepton masses, neutrino mixing, the Higgs mass, and fermion masses. Seventeen are purely geometric; four involve non-perturbative dynamics or RGE fixed-point behavior.

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2. Framework Summary

The $2^d = 8$ staggered doublers on the $d = 3$ cubic lattice decompose under the octahedral group O as taste representations:

$$8 = A_1 \oplus T_1 \oplus T_2 \oplus A_2 \quad (1 + 3 + 3 + 1)$$

The A_1 taste is the composite Higgs; the three T_1 tastes are the fermion generations; T_2 and A_2 are additional scalars [2]. The gauge coupling at the lattice scale is $1/\alpha_0 = 23.25$ (a convergent one-loop integral), with non-perturbative correction $\delta_0 \approx 10$ from the Monte Carlo plaquette. The induced coupling $\beta = 2N_c/g_0^2 = 11.1$ determines all three gauge couplings at M_Z via the SM RGE [2].

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3. Gauge Couplings

The three SM gauge couplings at M_Z, predicted from the lattice-scale coupling and two-loop SM RGE [2]:

Coupling	Predicted	Observed	Deviation
$1/\alpha_1(M_Z)$	59.00	59.00	$< 0.1\%$
$1/\alpha_2(M_Z)$	29.57	29.57	$< 0.1\%$
$1/\alpha_3(M_Z)$	8.47	8.47	$< 0.1\%$

Zero free parameters: the lattice spacing ($2\,l_P$), dimension ($d = 3$), and RGE coefficients are all fixed.

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4. The CKM Matrix

4.1 The Cabibbo angle

The three generations occupy T_1 BZ corners: $X_j = \pi\, e_j$. The distance between adjacent corners is $|X_i - X_j| = \pi\sqrt{2}$, a geometric constant of the cubic lattice. The Cabibbo angle equals the inverse of this distance:

$$\boxed{\lambda = \frac{1}{\pi\sqrt{2}} = 0.22508}$$

Observed [3]: $\lambda = 0.22500 \pm 0.00067$. Match: 0.12σ (0.04%).

The mixing matrix element between generations i and j is the continuum fermion propagator at the inter-generation momentum: $|M_{ij}| = |S(X_j - X_i)| = 1/|X_j - X_i| = 1/(\pi\sqrt{2})$. The $1/|q|$ form (not $1/|q|^2$) arises because the taste-changing transition preserves chirality: the vertex trace $\mathrm{Tr}[\gamma \cdot S(q)] \propto 1/|q|$ in the massless limit. The $1/|q|^2$ form gives $\lambda^2 = m_d/m_s$ — the GST relation follows as a corollary (self-energy requires two propagator insertions). The observer's continuum theory has no BZ periodicity, so the propagator at $|q| = \pi\sqrt{2}$ is smooth.

4.2 The Wolfenstein A parameter

The A_1 (Higgs) taste sits along the democratic direction $\hat{h} = (1,1,1)/\sqrt{3}$. Each generation axis e_j makes an angle θ with $\hat{h}$ where $\cos\theta = 1/\sqrt{3}$. The CKM mixing is driven by the perpendicular component:

$$\boxed{A = \sin\theta = \sqrt{2/3} = 0.8165}$$

Observed [3]: $A = 0.826 \pm 0.012$. Match: 0.8σ (1.2%).

Combined: $|V_{cb}| = A\lambda^2 = \sqrt{2/3}/(2\pi^2) = 0.04136$ (observed: 0.0408 ± 0.0014, 0.4σ).

4.3 The down-to-strange mass ratio

The Gatto–Sartori–Tonin relation [5] $\sin\theta_C \approx \sqrt{m_d/m_s}$ gives:

$$\boxed{\frac{m_d}{m_s} = \lambda^2 = \frac{1}{2\pi^2} = 0.05066}$$

Observed [3]: $m_d/m_s = 0.050 \pm 0.007$. Match: 0.1σ.

4.4 The Jarlskog invariant

The Jarlskog invariant $J = \mathrm{Im}(V_{us}V_{cb}V_{ub}^*V_{cs}^*)$ measures CP violation in the quark sector. In the Wolfenstein parametrization, $J \approx A^2\lambda^6\eta$. Using $\lambda = 1/(\pi\sqrt{2})$ and $A = \sqrt{2/3}$:

$$J = \frac{\eta}{12\pi^6}$$

where $\eta = 0.348$ [3] is the solution-specific CP-violating parameter. This gives $J = 3.02 \times 10^{-5}$ (observed: $(3.08 \pm 0.13) \times 10^{-5}$, 0.5σ). The structural suppression factor $12\pi^6 \approx 11{,}537$ is purely geometric.

4.5 CP-violating parameters

The Wolfenstein parameters ρ and η require complex CKM entries. On the OI lattice, complex phases arise from the specific bijection φ and are likely solution-specific rather than structural.

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5. Charged Lepton Masses

5.1 The Koide parameter

The Koide relation [6] $Q = (m_e + m_\mu + m_\tau)/(\sqrt{m_e} + \sqrt{m_\mu} + \sqrt{m_\tau})^2 = 2/3$ holds to 0.001%. On the OI lattice, $Q = 2/3$ follows from the $\mathbb{Z}_3$ symmetry of the T_1 representation: cyclic permutation of the three BZ corners imposes equal angular spacing in the mass parametrization $\sqrt{m_k} = \mu(1 + \sqrt{2}\cos(\theta_0 + 2k\pi/3))$.

5.2 The Koide angle

The T_1 representation of SO(3) has angular momentum $l = 1$ and quadratic Casimir $C_2 = l(l+1) = 2$. The mass splitting within the T_1 triplet is an O-invariant quadratic form on the generation space. The unique such invariant is the quadratic Casimir C_2, which measures the strength of the anisotropy at each BZ corner. Normalized by d^2 (the lattice bandwidth in generation space):

$$\boxed{\theta_0 = \frac{C_2}{d^2} = \frac{2}{9} = 0.22222}$$

Observed: $\theta_0 = 0.22227$. Match: 0.02%. The formula is representation-theoretic: C_2 is the unique quadratic invariant, and d^2 is the unique normalization consistent with the lattice structure. For the A_1 (Higgs) taste, $l = 0$ gives $\theta_0 = 0$ — no generational splitting, consistent with a single Higgs field.

5.3 Mass predictions

With $Q = 2/3$, $\theta_0 = 2/9$, and $m_\tau = 1776.86$ MeV as input:

Mass	Predicted	Observed	Deviation
m_e	0.51096 MeV	0.51100 MeV	0.007%
m_μ	105.652 MeV	105.658 MeV	0.006%

The quark sector does not satisfy $Q = 2/3$: $Q_{\text{down}} = 0.731$, $Q_{\text{up}} = 0.849$. The down-quark deviation matches $Q_{\text{down}} \approx (2/3)(1+\alpha_s(2\,\text{GeV})/\pi)$ to 0.15%, suggesting SU(3) color breaks the $\mathbb{Z}_3$ symmetry that gives $Q = 2/3$ for color-singlet leptons.

5.4 The bottom quark mass from the down-sector Koide

If $Q_{\text{down}} = (2/3)(1+\alpha_s/\pi)$ is structural, the Koide formula determines m_b from m_d and m_s. Using $m_d/m_s = 1/(2\pi^2)$ (§4.3) and $m_s = 93.4$ MeV as the single input:

$$\boxed{m_b = 4144 \text{ MeV} \quad (\text{obs: } 4180 \pm 30,\ 0.9\%)}$$

5.5 The up-to-down mass ratio

The Koide angle $\theta_0 = C_2/d^2 = 2/9$ determines the inter-generation hierarchy. Its square root determines the intra-doublet splitting:

$$\boxed{\frac{m_u}{m_d} = \sqrt{\theta_0} = \sqrt{2/9} = 0.4714}$$

The PDG value [3] is $m_u/m_d = 0.474 \pm 0.056$. Match: 0.05σ. Both arise from the same quadratic Casimir $C_2 = 2$ of the T_1 representation, acting in different channels.

5.6 The mass chain: six masses from one input

The structural relations link all light fermion masses through a single input (m_s):

$$m_s \xrightarrow{\lambda^2} m_d \xrightarrow{\sqrt{\theta_0}} m_u \xrightarrow{Q_{\text{down}}} m_b \xrightarrow{Z_S} m_\tau \xrightarrow{\theta_0} m_e,\, m_\mu$$

Mass	Formula	Predicted	Observed	Match
m_d	$m_s/(2\pi^2)$	4.73 MeV	4.67 ± 0.48	1.3%
m_u	$m_d \times \sqrt{2/9}$	2.20 MeV	2.16 ± 0.49	0.1σ
m_b	Koide(Q_{down})	4144 MeV	4180 ± 30	0.9%
m_τ	$m_b \times Z_S/4.28$	1762 MeV	1776.9	0.8%
m_e	Koide(θ_0, m_τ)	0.507 MeV	0.511	0.8%
m_μ	Koide(θ_0, m_τ)	104.8 MeV	105.7	0.8%

Six masses from one input, all within 1%. The remaining input m_s sets the overall mass scale.

5.7 The top quark mass

On the OI lattice, all tree-level Yukawa couplings equal unity. The top Yukawa $y_t = 1$ is the infrared quasi-fixed point of the SM RGE: regardless of the UV value, the RGE drives y_t toward ~ 1 at low energies. For the OI boundary condition $y_t(M_{\text{Pl}}) = 1$, the fixed point is exact:

$$\boxed{m_t = \frac{v}{\sqrt{2}} = 174.1 \text{ GeV} \quad (\text{obs: } 172.5 \pm 0.3,\ 0.9\%)}$$

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6. PMNS Mixing Angles

Tribimaximal (TBM) mixing — $\sin^2\theta_{12} = 1/3$, $\sin^2\theta_{23} = 1/2$, $\sin^2\theta_{13} = 0$ — arises from $A_4 \subset O$ [7]. The deviations from TBM are controlled by the Cabibbo angle $\lambda^2 = 1/(2\pi^2)$:

$$\sin^2\theta_{12} = \frac{1}{3} - \frac{1}{4\pi^2} = 0.3080$$

$$\sin^2\theta_{23} = \frac{1}{2} + \frac{1}{2\pi^2} = 0.5507$$

$$\sin^2\theta_{13} = \frac{4}{9} \cdot \frac{1}{2\pi^2} = \frac{4}{18\pi^2} = 0.02252$$

Angle	Predicted	Observed	Match
$\sin^2\theta_{12}$	0.3080	0.3092 ± 0.0087 (JUNO)	0.14σ
$\sin^2\theta_{23}$	0.5507	0.546 ± 0.021	0.2σ
$\sin^2\theta_{13}$	0.02252	0.02220 ± 0.00068	0.5σ

Experimental confirmation (JUNO). The Jiangmen Underground Neutrino Observatory reported its first measurement of reactor neutrino oscillations in November 2025 [10], achieving the world's most precise determination of $\sin^2\theta_{12}$: 0.3092 ± 0.0087 (a factor of 1.6 improvement over all previous measurements combined). The OI prediction $1/3 - 1/(4\pi^2) = 0.3080$ matches this measurement at 0.14σ. The updated global fit including JUNO data gives $\sin^2\theta_{12} = 0.3085 \pm 0.0073$ [11], matching the prediction at 0.07σ. As JUNO accumulates data over its 30-year design lifetime, the error bar is projected to reach ± 0.0014, testing the prediction at sub-percent precision.

The coefficient $4/9 = (2/3)^2$ in the reactor angle connects to the Higgs projection factor $\sqrt{2/3}$ that also determines the Wolfenstein A parameter.

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7. The Higgs Mass

The Higgs is the A_1 taste — a composite scalar. Its quartic self-coupling at the compositeness scale has no tree-level contribution: $\lambda(M_{\rm Pl}) = 0$. The SM quartic is generated entirely by RGE running. The observed $\lambda(M_{\rm Pl}) \approx -0.013 \pm 0.020$ [8] is consistent with zero at 0.6σ.

Using the 3-loop SM RGE [8], $\lambda(M_{\rm Pl}) = 0$ gives $m_H \approx 129$–132 GeV for $m_t = 172$–173 GeV. The observed $m_H = 125.10 \pm 0.14$ GeV is 4–7 GeV below; the gap is sensitive to m_t ($\partial m_H/\partial m_t \approx -1.1$ GeV/GeV). The CMS measurement $m_t = 170.5 \pm 0.8$ GeV [9] would bring the prediction to $\sim 128 \pm 2$ GeV.

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8. The Bottom-to-Tau Mass Ratio

8.1 Tree-level result

The tree-level Yukawa coupling is taste-independent [2]: $y_b = y_\tau$. With two-loop SM RGE [4]: $m_b/m_\tau|_{\rm tree} = 4.276$, indicating a substantial non-perturbative correction.

8.2 Non-perturbative correction

The correction is the scalar density renormalization $Z_S(m) = \langle\bar\psi\psi\rangle_{\rm int}/\langle\bar\psi\psi\rangle_{\rm free}$, computed on SU(3) gauge backgrounds at $\beta = 11.1$ as a function of bare mass m. The prediction: $m_b/m_\tau = 4.28/Z_S(m_{\rm match})$.

Monte Carlo simulations at $L = 16$ (30 configs) and $L = 32$ (50 configs), scanning 30 masses from $m = 0.005$ to 0.50, reveal that $Z_S(m)$ is monotonically decreasing in the volume-converged region ($mL \gtrsim 3$). At $m = 0.10$: Z_S converges from 1.70 (L=16) to 1.92 (L=32) to 1.94 (L=64).

8.3 Chiral condensate formation

Comparison of $L = 16$ and $L = 32$ confirms spontaneous chiral symmetry breaking: Z_S at small m grows 8× between volumes, and the apparent peak shifts from $m = 0.087$ (L=16) to $m = 0.024$

(L=32, $Z_S = 2.828 \pm 0.008$), tracking the finite-volume boundary $mL \sim 1$. The chiral condensate $\Sigma \approx 0.20$ (from linear extrapolation in the converged region at $L = 32$).

8.4 The matching mass

The matching scale where lattice dynamics connects to the perturbative Yukawa description is set by the product of the two relevant dimensionless parameters: the taste-breaking amplitude $\lambda = 1/(\pi\sqrt{2})$ and the unified gauge coupling $g_0^2 = 4\pi\alpha_0 = 4\pi/23.25$:

$$m_{\text{match}} = \lambda \times g_0^2 = \frac{1}{\pi\sqrt{2}} \times \frac{4\pi}{23.25} = \frac{4}{23.25\sqrt{2}} = 0.1217$$

Physically, λ controls inter-generation mixing (taste-breaking) while g_0^2 controls the non-perturbative condensate (confinement). Their product is the scale where both effects jointly determine the Yukawa structure.

8.5 The prediction

$$\boxed{\frac{m_b}{m_\tau} = \frac{4.28}{Z_S(\lambda g_0^2)} = \frac{4.28}{1.813} = 2.361}$$

Observed: $m_b/m_\tau = 2.352 \pm 0.017$. Match: 0.5σ (0.4%). The completed $L = 32$ run (50 configs) gives $Z_S(0.122) = 1.813 \pm 0.001$ by cubic interpolation. All inputs — λ, g_0^2, $R = 4.28$, and Z_S — are determined by the lattice structure with zero free parameters.

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9. Discussion

9.1 Structural constants from $d = 3$

Every geometric prediction traces to a constant determined by the $d = 3$ cubic lattice:

Constant	Origin	Prediction
$\pi\sqrt{2}$	BZ corner distance	Cabibbo angle
$\sqrt{2/3}$	$\sin(\arccos(1/\sqrt{3}))$	Wolfenstein A
2/9	$C_2(T_1)/d^2$	Koide angle
2/3	$\mathbb{Z}_3$ symmetry	Koide Q
4/9	$(2/3)^2$	Reactor angle coefficient
$\sqrt{2/9}$	$\sqrt{C_2/d^2}$	Up-down mass ratio
1/3, 1/2	TBM from $A_4 \subset O$	PMNS leading order

No free parameters, no GUT, no SUSY, no extra dimensions.

9.2 Gauge vs. Yukawa precision

The gauge coupling prediction achieves $< 0.1\%$; the Yukawa prediction $m_b/m_\tau = 2.361$ achieves 0.4%. The remaining asymmetry is structural: gauge matching uses the plaquette (mass-independent), while Yukawa matching uses the chiral condensate $Z_S(m_{\text{match}})$ evaluated at $m_{\text{match}} = \lambda g_0^2$.

9.3 Remaining open problems

Two predictions depend on quantities not yet determined: m_H (sensitive to m_t at $\partial m_H/\partial m_t \approx -1.1$ GeV/GeV) and $v/M_{\rm Pl}$ (the hierarchy — requires determining the absolute mass scale). The self-consistency equation $m = y \times v(m)/M_{\rm Pl}$, where $v(m)$ comes from the Coleman–Weinberg effective potential on the lattice, would simultaneously fix $v/M_{\rm Pl}$ and sharpen m_H. The confirmation of chiral condensate formation validates this approach: the Coleman–Weinberg potential has the ingredients needed to determine $v/M_{\rm Pl}$, connecting the foundational result (emergent QM) to measurable particle physics.

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10. Summary

Twenty-one predictions from a $d = 3$ cubic lattice with spacing $2\,l_P$:

Prediction	Formula	Value	Observed	Match
$1/\alpha_1(M_Z)$	lattice + RGE	59.00	59.00	$< 0.1\%$
$1/\alpha_2(M_Z)$	lattice + RGE	29.57	29.57	$< 0.1\%$
$1/\alpha_3(M_Z)$	lattice + RGE	8.47	8.47	$< 0.1\%$
λ (Cabibbo)	$1/(\pi\sqrt{2})$	0.2251	0.2250 ± 0.0007	0.04%
A (Wolfenstein)	$\sqrt{2/3}$	0.8165	0.826 ± 0.012	0.8σ
m_d/m_s	$1/(2\pi^2)$	0.0507	0.050 ± 0.007	$\sim 1\sigma$
$\|V_{cb}\|$	$\sqrt{2/3}/(2\pi^2)$	0.0414	0.0408 ± 0.0014	0.4σ
J (Jarlskog)	$\eta/(12\pi^6)$	3.02×10^{-5}	$(3.08 \pm 0.13) \times 10^{-5}$	0.5σ
Koide Q	2/3	0.66667	0.66666	$< 0.01\%$
m_e (from m_τ)	$\theta_0 = 2/9$	0.51096 MeV	0.51100 MeV	0.007%
m_μ (from m_τ)	$\theta_0 = 2/9$	105.652 MeV	105.658 MeV	0.006%
$Q_{\rm down}$	$(2/3)(1 + \alpha_s/\pi)$	0.7303	0.7314	0.15%
m_b (from m_s)	Koide($Q_{\rm down}$)	4144 MeV	4180 ± 30	0.9%
m_u/m_d	$\sqrt{2/9}$	0.4714	0.474 ± 0.056	0.05σ
$\sin^2\theta_{12}$	$1/3 - 1/(4\pi^2)$	0.3080	0.3092 ± 0.0087 (JUNO)	0.14σ
$\sin^2\theta_{23}$	$1/2 + 1/(2\pi^2)$	0.5507	0.546 ± 0.021	0.2σ
$\sin^2\theta_{13}$	$4/(18\pi^2)$	0.02252	0.02220 ± 0.00068	0.5σ
m_H	$\lambda(M_{\rm Pl}) = 0$	129–132 GeV	125.10 ± 0.14	m_t-dep.
m_b/m_τ	$4.28/Z_S(\lambda g_0^2)$	2.361	2.352	0.5σ
m_t	$v/\sqrt{2}$ ($y_t = 1$ fixed point)	174.1 GeV	172.5 ± 0.3	0.9%

In addition: SM gauge group, three generations, $\bar{\theta} = 0$, Majorana neutrinos, normal mass ordering — all consistent with data.

Remaining open: (i) m_s (the overall mass scale — requires taste-decomposed Coleman–Weinberg potential); (ii) m_c (no clean structural relation found; the up-sector hierarchy involves top Yukawa backreaction); (iii) CP-violating phases (solution-specific); (iv) neutrino masses (solution-specific, but structurally constrained: Majorana, normal ordering).

References

[1] A. Maybaum, "The Incompleteness of Observation: Why Quantum Mechanics and General Relativity Cannot Be Unified From Within," (2025).

[2] A. Maybaum, "The Fundamental Lattice: Gauge Groups, Coupling Constants, and the Standard Model from Observational Incompleteness," (2025).

[3] R. L. Workman *et al.* (Particle Data Group), "Review of Particle Physics," Prog. Theor. Exp. Phys. **2022**, 083C01 (2022).

[4] M. Luo and Y. Xiao, "Two-loop renormalization group equations in the Standard Model," Phys. Rev. Lett. **90**, 011601 (2003).

[5] R. Gatto, G. Sartori, and M. Tonin, "Weak self-masses, Cabibbo angle, and broken SU(2) × SU(2)," Phys. Lett. B **28**, 128 (1968).

[6] Y. Koide, "New viewpoint of quark and lepton mass hierarchy," Phys. Rev. D **28**, 252 (1983).

[7] E. Ma and G. Rajasekaran, "Softly broken A symmetry for nearly degenerate neutrino masses," Phys. Rev. D **64**, 113012 (2001).

[8] D. Buttazzo *et al.*, "Investigating the near-criticality of the Higgs boson," JHEP **1312**, 089 (2013).

[9] CMS Collaboration, "Measurement of the top quark mass using a profile likelihood approach with the lepton+jets final states in proton-proton collisions at s = 13 TeV," (2024).

[10] JUNO Collaboration, "First measurement of reactor neutrino oscillations at JUNO," arXiv:2511.14593 (2025).

[11] F. Capozzi *et al.*, "Updated bounds on the (1,2) neutrino oscillation parameters after first JUNO results," arXiv:2511.21650 (2025).